



中國海洋大學海德學院

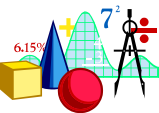
HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Statistical Modelling and Inference

Week 3, Lecture 2

Outline:

- i. Confidence intervals for one and two population proportions
- ii. Testing of hypotheses for one and two population proportions





Statistic

Parameter

Mean:

$\bar{X}$

estimates

μ

Standard
deviation:

s

estimates

σ

Proportion:

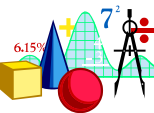
p_0

estimates

p

from sample

from entire
population





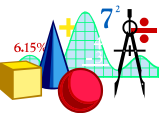
Standard Error

Quantitative Variable

$$SE(\bar{Y}) = \frac{S}{\sqrt{n}}$$

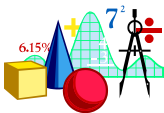
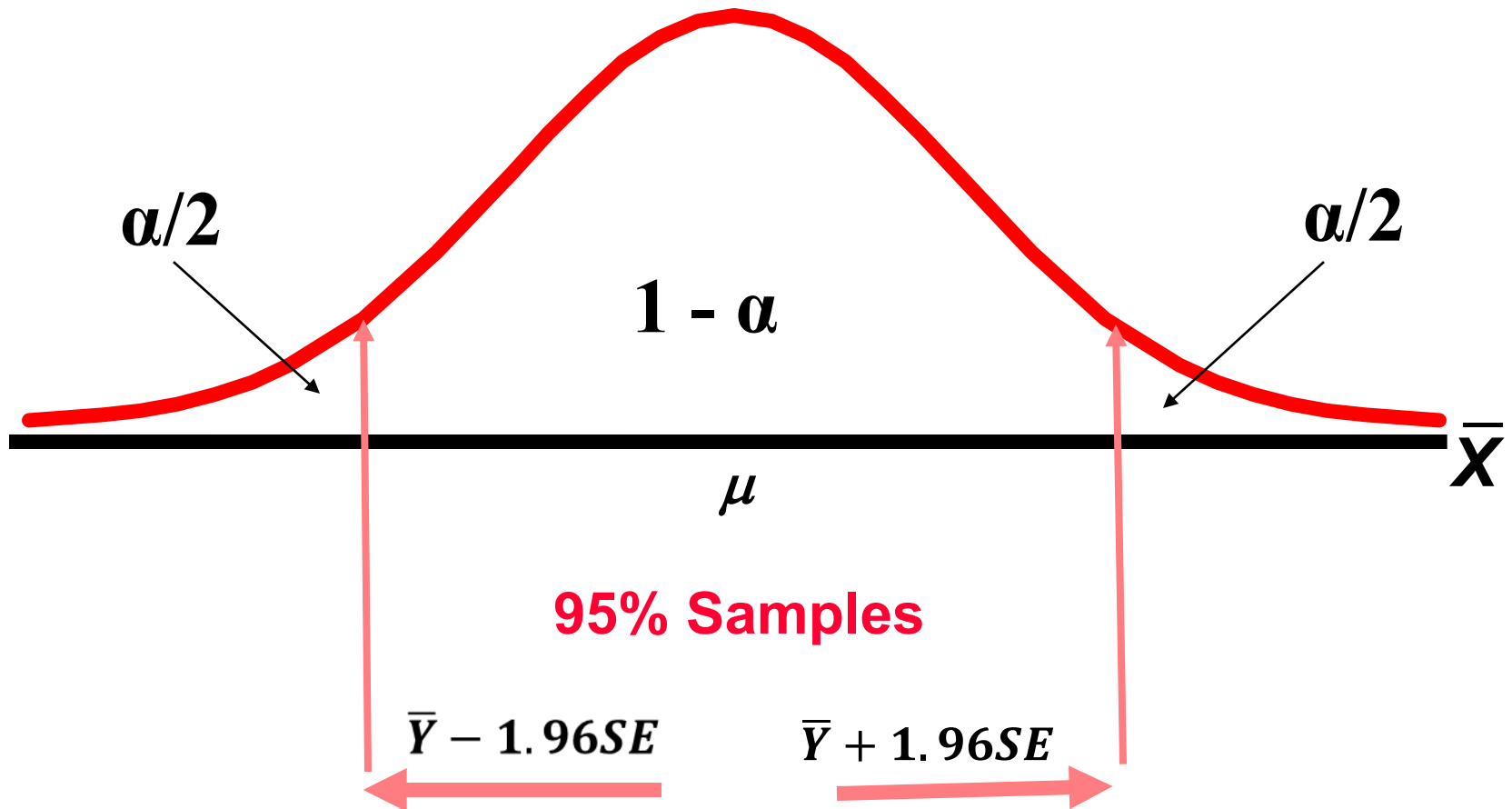
Qualitative Variable

$$SE(p) = \sqrt{\frac{p(1 - p)}{n}}$$



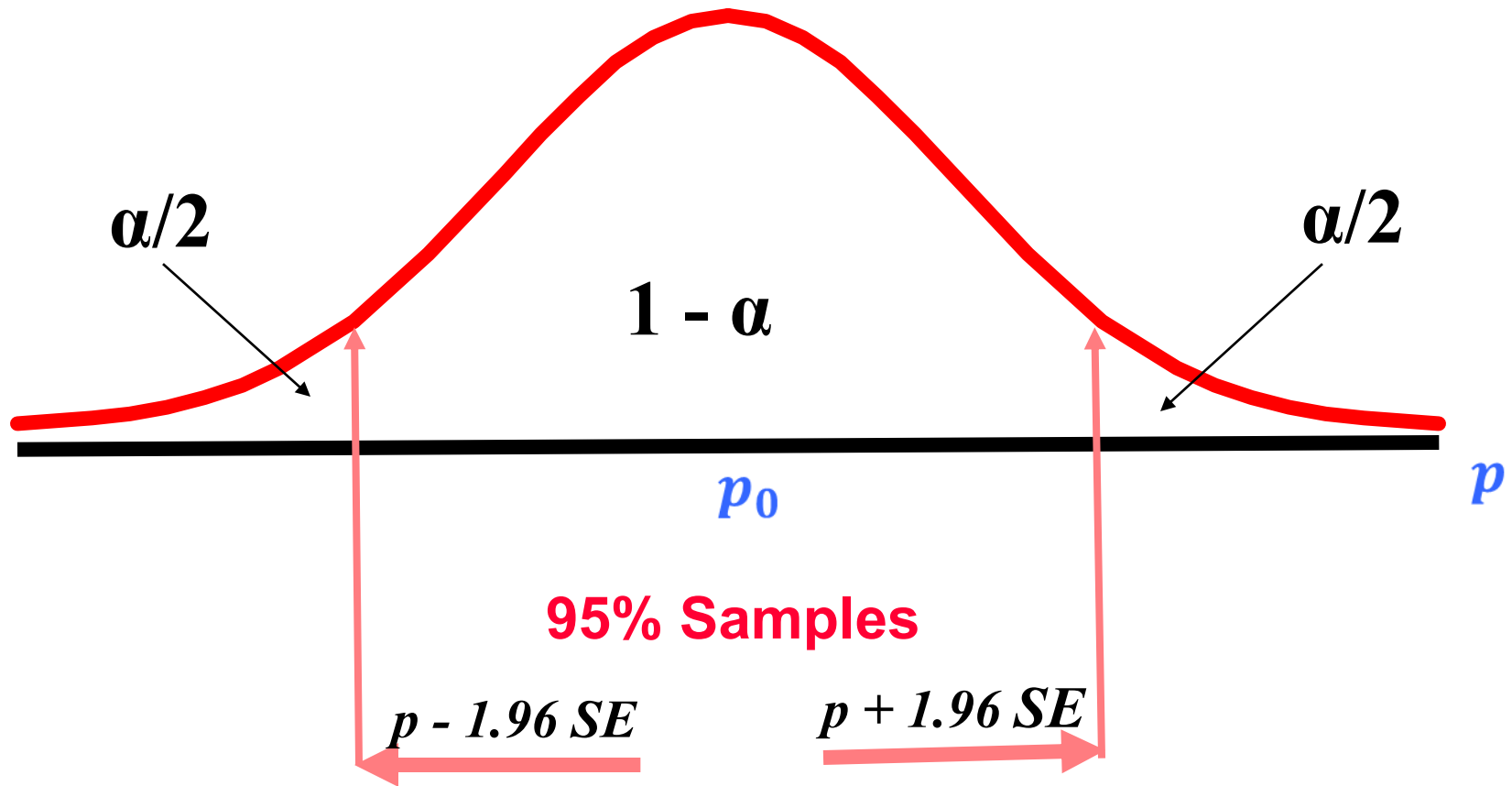


Confidence Interval for population mean μ





Confidence Interval for population proportion p





Example (CI for Proportion)

In a survey of 140 households, 35% had allergy to house dust.
Construct the 95% CI for the population proportion.

Given $n = 140$ and $p = 0.35$

For 95% CI for the population proportion, use $Z_{\frac{\alpha}{2}} = 1.96$

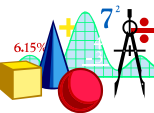
$$p \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}}$$

$$0.35 \pm 1.96 \sqrt{\frac{0.35(1-0.35)}{140}}$$

$$0.35 - 1.96 \times 0.04 \leq p_0 \leq 0.35 + 1.96 \times 0.04$$

$$0.27 \leq p_0 \leq 0.43$$

$$27\% \leq p_0 \leq 43\%$$





One-sample test for population proportion

Involves categorical variables fraction or % of population in a category sample proportion (p)

$$p = \frac{X}{n} = \frac{\text{number of successes}}{\text{sample size}}$$

Test statistic: (always Z test)

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Note: p_0 is proportion in population
(null hypothesis value)

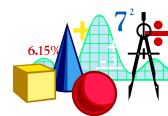
Use critical values: 1.96 at $\alpha=0.05$

2.58 at $\alpha=0.01$



Example (proportion)

- In a survey of diabetics in a large city, it was found that 100 out of 400 have diabetic foot. Can we conclude that 20 percent of diabetics in the sampled population have diabetic foot.
- Test at the $\alpha = 0.05$ significance level.





Solution

Hypothesis

$$H_0: p_0 = 0.20$$

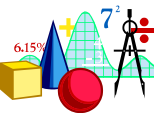
$$H_1: p_0 \neq 0.20$$

Test statistic:

$$Z = \frac{0.25 - 0.20}{\sqrt{\frac{0.20(1 - 0.20)}{400}}} = 2.5$$

Critical Value: at $\alpha = 0.05$ is 1.96

Conclusion: We have sufficient evidence to reject the H_0 value of 20%. We conclude that in the population of diabetic the proportion who have diabetic foot does not equal 0.20





For two population proportions

Suppose we have two populations:

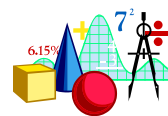
Population 1: proportion is p_1

Population 2: proportion is p_2

For samples:

Sample 1: size n_1 , successes x_1 and
proportion of success: $\hat{p}_1 = \frac{x_1}{n_1}$

Sample 2: size n_2 , successes x_2 and
proportion of success: $\hat{p}_2 = \frac{x_2}{n_2}$





Confidence interval for two Proportions

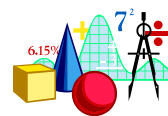
$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\frac{\alpha}{2}} \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\left(\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}\right)}}$$

Example:

A university wants to compare the effectiveness of two teaching methods A and B are online learning and traditional classroom. Based on samples data find 95% confidence interval for difference between two teaching methods.

Method A: $n_1 = 100$, $x_1 = 85$ and $\hat{p}_1 = 0.85$

Method B: $n_2 = 125$, $x_2 = 75$ and $\hat{p}_2 = 0.60$





Two-sample test for population proportion

Two-tailed test:

$$H_0: p_1 = p_2,$$

$$H_a: p_1 \neq p_2,$$

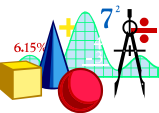
One tailed test:

right tailed: $H_0: p_1 \leq p_2$ and $H_a: p_1 > p_2$

left tailed: $H_0: p_1 \geq p_2$ and $H_a: p_1 < p_2$

Decision Rule: same as used in normal Z.

Or reject the null hypothesis if p-value $< \alpha$.





Two-sample test for population proportion

Test statistic:

Used pooled proportion under H_0 :

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

$$Z_{obs} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$



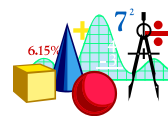
Two-sample test for population proportion

Example:

Time magazine reported the result of a telephone poll of 800 adult Chinese. The question posed of the Chinese who were surveyed was: "Should the federal tax on cigarettes be raised to pay for health care reform?" The results of the survey were:

Non-Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
$x_1 = 351$ (said yes)	$x_2 = 41$ (said yes)
$\hat{p}_1 = 0.58$	$\hat{p}_2 = 0.21$

Is there sufficient evidence at 5% level of significance, to conclude that the two populations smokers and non-smokers differ significantly with respect to their opinions?





**Thank
You**

谢谢

